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REVIEW ARTICLE

Three-Dimensional Methods for Quantification of Cancellous Bone Architecture

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Recent development in three-dimensional (3-D) imaging of cancellous bone has made possible true 3-D quantification of trabecular architecture. This provides a significant improvement of the tools available for studying and understanding the mechanical functions of cancellous bone. This article reviews the different techniques for 3-D imaging, which include serial sectioning, X-ray tomographic methods, and NMR scanning. Basic architectural features of cancellous bone are discussed, and it is argued that connectivity and architectural anisotropy (fabric) are of special interest in mechanics–architecture relations. A full characterization of elastic mechanical properties is, with traditional mechanical testing, virtually impossible, but 3-D reconstruction in combination with newly developed methods for large-scale finite element analysis allow calculations of all elastic properties at the cancellous bone continuum level. Connectivity has traditionally been approached by various 2-D methods, but none of these methods have any known relation to 3-D connectivity. A topological approach allows unbiased quantification of connectivity, and this further allows expressions of the mean size of individual trabeculae, which has previously also been approached by a number of uncertain 2-D methods. Anisotropy may be quantified by fundamentally different methods. The well-known mean intercept length method is an interface-based method, whereas the volume orientation method is representative of volume-based methods. Recent studies indicate that volume-based methods are at least as good as interface-based methods in predicting mechanical anisotropy. Any other architectural property may be quantified from 3-D reconstructions of cancellous bone specimens as long as an explicit definition of the property can be given. This challenges intuitive and vaguely defined architectural properties and forces bone scientists toward 3-D thinking. (Bone 20:315–328; 1997) © 1997 by Elsevier Science Inc. All rights reserved.

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Introduction

The mechanical properties of cancellous bone have major socioeconomic importance. Osteoporosis is often considered a disease of cancellous bone, cancellous bone has been related etiologically to osteoarthrosis, and long-term success of orthopedic implants depends on a sound cancellous bone stock. The ultimate aim of most cancellous bone research is therefore to understand the factors that determine the mechanical properties, how these properties are maintained, and how bone reacts to changes in its environment.

The compressive test is the standard technique for studying mechanical properties of cancellous bone, and many reports of compressive strength and Young's modulus have been published.[25,48,92,119] There are, however, two important problems with all studies of mechanical properties of cancellous bone. First, the mechanical properties have almost never been described completely. The mechanical properties vary with loading orientation[41,103,168]; that is, cancellous bone is mechanically anisotropic, and few investigators have tried to describe this variation. The scarcity of these studies is caused by the complexity of the problem. A meticulous description of anisotropic elastic properties may, in the general case, involve up to 21 elastic constants and, using a traditional experimental approach, this description is virtually impossible. Second, reported mechanical constants are only rarely accurate. All mechanical tests are influenced by inherent errors and problems, which include specimen geometry,[90,91,177] friction at endplates,[14,20,109] structural end phenomena,[93,132,188] storage,[112,156,163] continuum assumption,[65] viscoelasticity,[113] and temperature effects.[13] The compressive test may seem simple, but traditionally determined stiffness and strength may be wrong by up to 40%,[136] and worse still, the inaccuracy will probably vary with the trabecular architecture. Traditional mechanical testing of cancellous bone may, consequently, not be able to detect changes in mechanical properties.

In the second half of the 19th century Meyer[122] and Wolff[186,187] observed that cancellous bone has "a well-motivated architecture, which is closely related to its statics and mechanics," and it was suggested that trabeculae align along stress trajectories; that is, orientations where only pure compressive or tensile stresses occur. This theorem (Wolff's law) has the important corollary that the architecture of cancellous bone determines its mechanical properties. One remedy to the problems of mechanical tests would be to use the corollary of Wolff's law, and study changes in trabecular architecture instead of changes in mechanical properties. It is, however, a prerequisite that the architectural variables relate to the mechanical properties.

Density is the one architectural variable, which has been studied most intensely. For single anatomic regions and orientations, most elastic properties and strength are well predicted by density,[15,78,103,149] which is expected from theoretical studies.[3,179] The only elastic constant, that seems independent of density, is Poisson's ratio.[103,133] If different species, individuals, anatomic

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regions, or orientations are compared, one equation cannot fully describe the density–mechanics relation.[48,77,103,110,123] The same effect may be seen with fluoride treatment.[102,124,154] The variation has been explained by differences in trabecular architecture independent of density.

In an attempt to study principles of cancellous bone micromechanics, but avoiding complex real structures, several idealized models have been suggested.[121,146,168,185] Good agreement between models and mechanical properties of industrial foams has been observed, but the agreement for cancellous bone has been less successful.[5] The reason may be the large variation in trabecular architecture, which makes the use of idealized models difficult, but verification of the models also poses various problems.

There is, except for density, no a priori knowledge about which architectural aspects should be included in a study of architecture–mechanics relations of cancellous bone. But there seems to be at least two obvious aspects that should be studied: connectivity and anisotropy (**Figure 1**). Compared with these characteristics, many other architectural features seem secondary: mean trabecular volume, slenderness of individual trabeculae, curvature of trabeculae, etc. All of these aspects require 3-D information, and some of the properties have to be examined in 3-D space.

Traditional histomorphometry is based on 2-D sections, but for various reasons, which will be discussed in what follows, many of the methods are inadequate for describing cancellous bone architecture. Recent development in 3-D imaging of cancellous bone has, however, made the direct study of 3-D architecture and mechanics possible. This provides for a wealth of possibilities and unbiased determination of central architectural properties. The purpose of this article is to review this recent development in 3-D quantification of cancellous bone architecture.

In much of the literature, the adjectives *cancellous, trabecular,* and *spongy* are used interchangeably. Throughout this article the term *cancellous bone* will be restricted to the three-dimensional lattice-like structure composed by multiple trabeculae, and the term *trabecular bone* will be used to denote the material that constitutes individual trabeculae.[142] The term *spongy bone* will not be used.

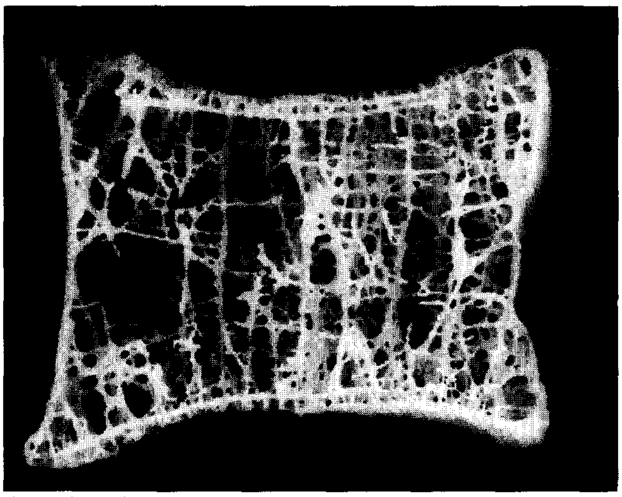

Figure 1. Serially reconstructed 8-mm-thick section through an osteoporotic vertebral body.[139] The resolution is 40 μm. The characteristic architectural features of cancellous bone are easily identified: multiple connections (trabeculae) and anisotropy (preferred orientations of the trabeculae).

3-D Reconstruction

Three-dimensional reconstruction of biological tissues is a classic method, which goes back to at least the middle of the 19th century.[44] Early 3-D reconstructions were based on tedious and time-consuming routines, which did not allow routine applications, but new computerized methods have significantly changed this.

Serial Reconstruction

The first 3-D reconstruction of cancellous bone seems to have been that done by Amstutz and Sissons,[2] who used serial sectioning to reconstruct regions from a vertebral body. The result was models built from plastic, which were used for qualitative descriptions and quantification of volume fraction and surface density. The method used by Amstutz and Sissons is extremely time consuming. Each individual section has to be photographed and reproduced onto a sheet of paper or other material. This is done for a number of sections with known distance, and the entire set of reconstructed sections then have to be realigned, so that a line perpendicular to the sections is passing through the same *x–y* coordinate in each image. Various techniques have been suggested for manual[44] and automated[79] realignment, but it remains a tedious and error-prone process.

The introduction of the automated serial sectioning technique[6,135,139] solved some of the problems of serial sectioning. Instead of using individual slices, the cancellous bone specimen is embedded in black epoxy, and the block surface itself is used for imaging, which effectively abolishes distortion artifacts. The position of the embedded block is precisely controlled at the time of image grabbing, which makes realignment an inherent quality of the method. With the present implementation of automated serial sectioning, close to 600 sections can be produced per hour with a resolution of 1/1000 of the image field.[139] See Figure 1 for an example.

X-Ray CT Methods

Various methods have been described for producing 3-D reconstructions of cancellous bone using X-rays. In conventional X-ray computerized tomography (CT), a linear array of detectors record projections of X-ray attenuation, and from numerous projections a 2-D image may be calculated.[67] The classic clinical use of CT images is qualitative, but making use of the densities in a CT image yields valuable information and has become known as quantitative computerized tomography (QCT). Recent development using this technique has resulted in images with high 3-D resolution, which may be used for 3-D reconstruction of cancellous bone regions.[127]

Instead of using a 1-D array of detectors, one may have a 2-D array, and from the attenuation information in a number of projections a 3-D image may be calculated. Feldkamp was a pioneer with this technique, and he developed what has become known as the μ-CT scanner.[35,37] The μ-CT scanner employs an X-ray point source in a cone-beam design,[34] and the resolution limitations of this system are among other factors determined by the image intensifier and the diameter of the point source. As with any QCT technique based on polychromatic X-rays, beam hardening artifacts occur.

The X-ray tomographic microscope (XTM)[9,10,95] is also based on a number of 2-D projections of a specimen, but the technique has important modifications compared to the μ-CT scanner. These modifications have become possible through the use of synchrotron radiation. First, the radiation is monochromatic, that is, no beam hardening; second, the radiation is parallel as opposed to the cone-beam design, which reduces certain image distortions;[98] and third, the X-ray beam is of high intensity. Specific advantages include a higher spatial resolution and lower radiation dose and time per scan. The highest resolution published for cancellous bone is about 8 μm,[10] but an even better resolution should be obtainable.[10,95,96] The XTM has very interesting possibilities of in vivo scanning, which allows sequential scanning of the same cancellous bone region.[97,98]

NMR Imaging

Recent studies have shown that nuclear magnetic resonance (NMR) imaging may also provide a sufficiently high resolution for 3-D reconstruction of trabecular architectures.[17,80,88,115] This has interesting perspectives in noninvasive determination of architectural properties of cancellous bone.

Segmentation

No matter which of the above methods is used for 3-D reconstructing a cancellous bone specimen, the output from the method is a 3-D gray-level image, typically with 256 gray levels. This information has to be reduced, so that each voxel has either the value "bone" or the value "marrow." This process is called segmentation, and various techniques have been used.[37,106,114,127,135,139]

Connectivity

There is hardly one publication about cancellous bone architecture which does not emphasize the branching 3-D nature of plate- and rod-like trabeculae; there are two interesting aspects in this. First, except for two studies,[57,137] there has probably not been given an explicit definition of trabecula, not to mention a plate or rod. Second, all but a few recent studies have aimed at cancellous bone connectivity in single two-dimensional sections only, which is theoretically impossible.

Qualitative Observations

Following the classic descriptions of cancellous bone architecture[99,122,186,187] more than half a century went by before new progress was seen. Whitehouse provided detailed descriptions of cancellous bone architecture using SEM,[182,183] and many later observations have been reported using various techniques.[32,62,82,125,157] Generally, architectures have been described using terms like "clearly connected" and "obviously disconnected." Except for overt pathologic conditions, the limited changes, which probably occur in a wealth of different pathologic and normal conditions, may not be described using qualitative methods.

Surrogate Measures

In an attempt to quantify connectivity, various methods have been suggested using individual 2-D sections. The first of these attempts seems to have been by Pugh et al.[147] using the *trabecular contiguity ratio.* This concept was developed to describe the degree of contact between particles of granular materials.[58] Applying the principle to cancellous bone is not straightforward, because trabeculae are not isolated particles, and the application requires counting "missing" trabeculae,[147] which may be even more difficult than counting existing trabeculae.

A set of measures based on skeletonization has been developed.[18,19,42] Results are expressed by a series of ratios between nodes, termini, cuts, ends, and loops. For all skeletonization procedures, one should be aware that skeletons produced may be homotopic (same topology), but they depend on the algorithm used. Worse, however, the measures are purely 2-D and have no known relation to 3-D connectivity, as discussed in what follows.

Counting the number of separate profiles has been undertaken by a few investigators.[7,66,73,74] This method has the absurd consequence that marrow and bone connectivity may turn out to be different,[73,74] and that connectivity may depend on the orientation of the section.[7] Connectivity is a scalar and cannot depend on orientation, and the connectivity of bone and marrow must be identical.[4,137] Another suggested method consists of counting individual trabeculae, and ad hoc rules have been made for identifying junctions and for determining when a trabecula is long enough to be identified as such.[1,76]

Separate measures of horizontal and vertical trabeculae using intersection counts with vertical and horizontal lines have been suggested,[82] and so have closely related pseudo-3-D measures.[31] The units of these measures are $length^{-1}$. All of these measures require subjective assessments in defining a trabecula, and even more so if classification into rod-like and plate-like trabeculae is attempted.[31]

A trabecular bone pattern factor (TBPf) has been suggested,[61] which is claimed to be a measure of connectivity. The measure may seem related to the curvature of a 2-D structure and as such to the connectivity,[29] but the insinuated relation is nonexisting. Take two circular structures as an example, and let one structure have a radius of 1 and the other a radius of 2. These structures will both have a connectivity of zero. If the dilation[61] increases the radius by 0.1, then the structures will have a TBPf of 0.95 and 0.49, respectively. This example illustrates clearly that TBPf has no relation to connectivity, not even in 2-D space.

Some of the investigators who have used these 2-D methods have stated that no attempt has been made to venture into 3-D space, or that the connectivity in a random section relates to the connectivity of the 3-D structure. One cannot argue with the first of these two statements except for questioning the relevance of a pure 2-D measure. As for the latter statement, no matter how any of the 2-D measures may seem to be intuitively related to 3-D connectivity, no relation exists between connectivity in a random 2-D section and connectivity in 3-D space.[30,47,57] For specific purposes, any of these surrogate measures may show significant differences between different groups, but no one can be sure exactly what is being measured. As a consequence, the surrogate 2-D measures will not be considered further.

Star volume gives an unbiased measure of the marrow volume, which may be seen unobscured from a typical point in marrow.[176] It was argued that this measure is related to connectivity, because disappearance of trabeculae (i.e., decreased connectivity) would result in an increased marrow star volume. Increases in connectivity (e.g., perforations of plate-like trabeculae) may, however, also increase the star volume. Consequently, there is no unequivocal relation between connectivity and star volume, which consequently remains a surrogate measure of connectivity.

The Model-Based 2-D Approach

Parfitt et al.[141] introduced the plate model of cancellous bone. In this model it is assumed that trabeculae are parallel, infinite plates, and that a 2-D section with random orientation is available for analysis. On the 2-D section the trabecular volume fraction, V_V (also called BV/TV), and the trabecular surface per volume,

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S_V (also called BS/TV), are measured using conventional techniques.[173,180] Assuming the plate structure, various parameters may be calculated from the two measured quantities.[141] These measures include the mean trabecular plate density (MTPD), the mean trabecular plate thickness (MTPT), and the mean trabecular plate separation (MTPS). The first of these measures is the mean number of plates traversed by a line of unit length perpendicular to the plates, and the unit of MTPD is *length*$^{-1}$. MTPD is a measure of the trabecular density, and is as such an attempt at quantifying connectivity. The calculation of MTPT is identical to that proposed by Whitehouse.[183]

The introduction of the plate model has been of value for both enhancing the perception of cancellous bone structures and for providing quantitative information about normal and pathologic trabecular architectures: see, for instance, Kleerekoper et al.[101] The model assumption has, however, the consequence that researchers should be very careful to verify the assumption before interpreting results. Only rarely has the application of the method been refrained from, because the architecture has been found non-plate-like.[97]

One classic example of the use of the model is the work by Kleerekoper et al.[101] The results of Kleerekoper's article are often summarized by saying that a certain volume of trabeculae distributed in thick, widely spaced trabeculae is inferior to the same volume distributed in many thin trabeculae. What Kleerekoper et al. did show was that there was a difference in MTPD and MTPT between two populations (osteoporotics and normals), but this may only be interpreted in the way mentioned if the architecture in both populations was plate-like. Checking the model assumption was not done, and doing so would be a nontrivial task.

This objection is important for two reasons. First, the way one conceives the results of a study depends very much on the translations of MTPD, etc. into words. If a study suggests differences in MTPT the next step may be to design studies using other techniques to further characterize the observed difference, and the design of these studies will ultimately be determined by the conception of MTPT. Second, observed differences in, say, MTPT may not be caused by changes in *trabecular thickness*, but may be caused by changes in the architecture, which affects the model. This is a classic problem of model-based measures. Alternative model-based approaches exist,[8,116] and these models carry the same fundamental problems.

In 1987, plate-model-derived parameters were given a new nomenclature,[142] and one of the intentions was to provide descriptive names without implicit assumptions. The most important change was that the mean trabecular plate density was renamed *trabecular number*. The new nomenclature left out the word "plate," giving the illusion that MTPD is independent from the plate model and, more important, it directly related MTPD to connectivity. The number of trabeculae is solidly related to connectivity (see subsequent text), and this relation is by no means provided by MTPD (or "trabecular number," Tb.N).

The Topological Approach

A topological approach[4,29,59,60] to cancellous bone connectivity solves the problems encountered in the model-based approach.[57,137] For two-phase (bone and marrow) 3-D structures, the topological properties of interest are the number of connected components (particles) of the two phases, and the connectivity, which is identical for the two phases.

A *tree* is, in a topological sense, a node-and-branch network in which only one path exists between any two nodes. If an additional branch is added between two nodes, then more than one path will exist between a number of node pairs, and the node-and-branch network is now *multiply connected.* If a branch is cut in a tree, then the network will be separated into two parts, but a number of branches may be cut in a multiply connected network without separating the network. *Connectivity* may be defined as the maximal number of branches that may be cut without separating the structure. If cancellous bone is viewed as a node-and-branch network, then the connectivity is the number of trabeculae minus one, and this defines *a trabecula* in a topological sense.[137]

The *Euler number,* χ, is the key to all determinations of connectivity, and it is related to connectivity and the number of bone and marrow particles:

$$\chi = \beta_0 - \beta_1 + \beta_2 \tag{1}$$

where β_0 is the number of bone particles, β_1 is the connectivity, and β_2 is the number of marrow cavities fully surrounded by bone. The Euler number may easily be calculated from the voxel-based data set of a 3-D reconstruction.[137]

There seems to be consensus that the trabecular bone phase is one fully connected structure without isolated parts ($\beta_0 = 1$), and that no marrow cavities exist fully surrounded by bone and separated from the main marrow space ($\beta_2 = 0$). This leads to the following simplification of equation (1)[137]:

$$\chi = 1 - \beta_1 \tag{2}$$

Recently, isolated trabecular parts have been demonstrated in a rat osteoporosis model,[98] but from a mechanical point of view, these parts are of no significance. Fracture healing and bone formation are other obvious exceptions to the topological assumption expressed in equation (2), but one can get information about the connectivity of these exceptions by determining χ, β_0, and β_2 separately and using equation (1).

In the segmentation process of any 3-D reconstruction, noise will result in isolated "bone" and "marrow" particles, which will lead to a violation of the topological assumption expressed in equation (2). The isolated bone and marrow particles may be removed in a purification step.[79,107,137]

The edge problem needs special attention in all quantifications of connectivity.[137] The Euler number of the union of two sets A and B is given by[59]:

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B) \tag{3}$$

Neglecting equation (3) has the serious implication that the Euler number of a bone specimen will not equal the sum of the Euler number of parts of the specimen. This means that, if one wants to determine the connectivity density in a region by cutting out a specimen from this region, then an incorrect connectivity density will be determined depending on the specimen size and the topology of the specimen surfaces. The magnitude of this error is unpredictable, except that the error tends to decrease as the examination volume increases. A solution to the problem is based on equation (3) and is described in detail elsewhere.[137]

Practical determinations of connectivity may be based on 3-D reconstructions, in which case the Euler number is determined as indicated previously.[137] Another option, which was developed in parallel to the 3-D solution, is based on the disector principle.[57] In this method, the Euler number change is determined for pairs of 2-D sections and, from knowledge of the fraction examined, the Euler number of the entire specimen may be estimated. A method related to this has been used by Boyce et al.,[11] and it has been demonstrated that MTPD (Tb.N) may decrease, while connectivity increases.[12]

Connectivity vs. Density

Goldstein et al.[49,50,54] have suggested that a constant number of trabeculae will exist for a given density, but the calculations did not take into account the edge problem, even though the problem was briefly discussed. It was concluded that adjusting the Euler number for the edge problem would probably not significantly influence the conclusion, but this assumption is unsubstantiated. An additional problem with the Goldstein data is that positive values of the Euler number were observed, which may point to a purification problem; cf. equation (1).

Others have also found connectivity to be positively correlated with density,[37,97,98,137] but in several of these observations there is considerable spread around the regression lines. Instead of explaining this spread by experimental errors, there are indications that it may reflect true differences in trabecular architecture.[108,137]

The Size of a Trabecula

Several methods have been described to quantify *trabecular thickness*,[43,141,178] and much attention has been paid to it. Considering the complex architecture of cancellous bone and the difficulty of defining thickness, it seems strange that trabecular thickness has become such a popular measure. The fact that histomorphometry has traditionally been performed on 2-D sections may explain the interest in "thickness." As stated by Wakamatsu and Sissons in 1969—before reluctantly quantifying "thickness"—"The arrangement of the bony structures is such that it is not easy to define, or to measure, what is commonly referred to as 'trabecular thickness.' "[178]

A more obvious measure of mean trabecular size is the mean volume of an individual trabecula, which has become accessible with the unbiased connectivity measures. If the specimen volume, V, the trabecular volume fraction, V_V, and the connectivity, β_1, are known, then the *mean trabecular volume*, MTV, is given by:

$$\text{MTV} = \frac{V \cdot V_V}{\beta_1 + 1} \tag{4}$$

A measure of the mean spacing between trabeculae would be the average size of a "marrow loop." The number of marrow loops equals the trabecular connectivity,[137] so the *mean marrow space volume*, MMSV, may be calculated as:

$$\text{MMSV} = \frac{V \cdot (1 - V_V)}{\beta_1 + 1} \tag{5}$$

where V_V still is the bone volume fraction.

Anisotropy

The most striking, architectural property of cancellous bone is probably its anisotropy, that is, orientation of trabeculae (see Figure 1). The mechanical properties of bone are also anisotropic,[41,103,118,168] so the first requirement for a formulation of a relation between mechanics and architecture must be an ability to handle anisotropy.

A brief description of anisotropy terms and types is needed for the following discussion. The term *orientation* denotes the hemispherical orientation of an axis as opposed to the spherical *direction* of a vector.[38,117] Let α be a property of interest. If α does not change with orientation, then the material is *isotropic* with respect to α. If α is considerably larger (or smaller) in one orientation, ω, than in any other orientation, and if α is constant for rotation around ω, then the material is *transversely isotropic* with respect to α. If there is a main orientation ω, and α has a maximum and a minimum in a plane perpendicular to ω, then the material is *orthotropic* with respect to α. For a detailed discussion of anisotropy types see Cowin and Mehrabadi.[24] The mathematical term *second-rank tensor* will be used in what follows, and it may be thought of as a mathematical operator. The quadratic form of a second-rank positive definite tensor is the mapping of a sphere onto an ellipsoid, and the reader may think of an ellipsoid when the term fabric tensor is used in what follows. The *eigenvectors* of the tensor give information about the direction of the axes of the ellipsoid, and the *eigenvalues* express the radii of the ellipsoid. Any textbook in linear algebra will cover the mathematics used in this section.

Orientation-dependent measures in 3-D space may be expressed by a sample of unit vectors, and one may express the properties of the sample by parameters of statistical models.[38,117] Kanatani[85] took a different approach, in which he expressed the properties of a sample of unit vectors by a series of even-ranked tensors. This approach may be compared to fitting a polynomial to a set of points on an x–y plot; the polynomial can be made to go through every single point if a sufficiently high degree is chosen. Kanatani[85] gave the name *fabric tensors* to the individual even-ranked tensors. Oda[131] found that the second-rank fabric tensor gave good predictions of mechanical properties of selected materials.

Cowin introduced the term *fabric tensor* in bone mechanics, and he defined a fabric tensor as any positive definite, second-rank tensor that gives a local description of the architectural anisotropy (also called *fabric*).[21,22,169] The general nature of this definition should be noted. Cowin also developed formulations of the dependency of mechanical anisotropy upon fabric,[21,23] which were based on the assumptions that fabric tensor main orientations correspond to mechanical main orientations and that the material properties of trabecular bone are isotropic. An important result of Cowin's work is that the type of material anisotropy (orthotropy, transverse isotropy, isotropy) will manifest itself in the eigenvalues of the fabric tensor. Different fabric tensors have been proposed for Cowin's constitutive anisotropy relations.

The Mean Intercept Length Method

Whitehouse[183] and Raux et al.[148] provided the first detailed descriptions of cancellous bone architectural anisotropy using an intersection counting technique. Raux et al. expressed the results as *roses of orientation*,[173] whereas Whitehouse took results one step further and expressed the results by the *mean intercept length* (MIL) measure.

The principle of the MIL method is to count the number of intersections between a linear grid and the bone/marrow interface as a function of the grid's 3-D orientation, ω[173,180]; see **Figure 2A**. The mean intercept length (the mean length between two intersections) is simply the total line length divided by the number of intersections:

$$\text{MIL}(\omega) = \frac{L}{I(\omega)} \tag{6}$$

Whitehouse[183] defined the mean intercept length in bone (MIL_b) and marrow (MIL_m) by

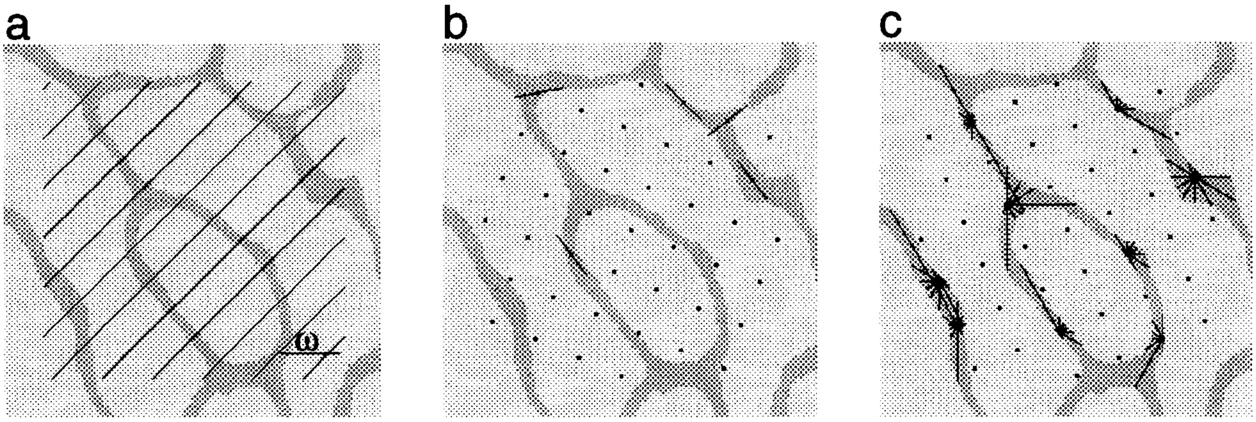

Figure 2. Principles for determining architectural anisotropy. (a) Mean intercept length (MIL) measurement. For each orientation, ω, of a randomly translated linear grid the number of intersections between the grid and the bone/marrow interface is determined. (b) Volume orientation (VO) measurement. A randomly translated point grid is placed on the structure. For each point hitting the phase of interest (bone) the orientation of the longest intercept through the point is determined, and this orientation is the local volume orientation for the point. In the figure, five local volume orientations are sampled and depicted by the compass needles. (c) Star volume distribution (SVD) and star length distribution (SLD) measurements. A randomly translated point grid is placed on the structure. For each point hitting the phase of interest (bone) the intercept length through the point is determined for several orientations. The intercept lengths are used directly for SLD and cubed for the SVD measure. In the figure intercept lengths have been determined for six orientations.

$$\mathrm{MIL}_p = V_V(p) \cdot 2 \cdot \mathrm{MIL}(\omega) \tag{7}$$

where p is the phase of interest (bone or marrow) and V_V is the volume fraction. The reason for the factor 2 is that 2 · MIL is the mean length of a bone + marrow intercept. Multiplying MIL by a constant as in equation (7) does not change the anisotropy information. The definition in equation (7) has been used by some investigators,[54,105,161,162] and their argument for doing so is that the absolute size of intercept lengths (or pore size) may be of significance in architecture–mechanics relations. There are, however, empirical and theoretical arguments for disregarding pore size,[45,46,169,179] and benefits of using equation (7) have yet to be proven.

Whitehouse[183] observed that 2-D MIL results, when plotted in a polar diagram, generate an ellipse, and the parameters of this ellipse would thus provide a convenient way of quantifying MIL anisotropy. These 2-D observations may be generalized to 3-D space, where a polar plot of the MIL data is assumed to approximate an ellipsoid,[184] and an ellipsoid may be expressed by the quadratic form of a second-rank tensor, known as the *MIL fabric tensor*.[22,64]

When 3-D reconstructions are at hand, one may determine the MIL data directly in 3-D space.[140] Methods for determining the MIL fabric tensor from perpendicular 2-D sections have been described.[64,87] Hodgskinson and Currey[73,74] determined MIL ellipses on the four faces parallel to the loading orientation on cancellous bone cubes and calculated a combined expression of the anisotropy of the four faces. A similar approach was taken by Harrigan et al.[63] The latter approaches are empirical, and a direct relation to Cowin's "constitutive" structure–property relations does not exist.

MIL results depend entirely on the interface between bone and marrow.[71,72,86,143] This has the somewhat surprising effect that, obviously anisotropic structures may appear isotropic when examined with the MIL method[39,134,138] (see Figure 4).

The Volume-Based Methods

Because MIL is unable to detect some forms of architectural anisotropy, volume-based measures were introduced with the *volume orientation* (VO) method.[134] The results of using the volume-based measures may also be expressed as positive definite, second-rank tensors, and consequently they are competitors for the position of providing the best fabric tensor for Cowin's structure–mechanics relations. The volume-based methods shift the interest of architectural anisotropy from interface to volume.

Volume orientation. A *local volume orientation* is defined for any point within a trabecula as the orientation of the longest intercept through the point[134] (see Figure 2B). The local volume orientation, ω, is in 2-D space a semicircular orientation, in 3-D space a hemispherical orientation, and it is assumed that ω is uniquely defined for any point within bone. The result of volume orientation (VO) measurements may be expressed by a *VO fabric tensor*,[140] which is a description of the typical distribution of trabecular bone volume around a typical point within a trabecula. This is in contrast to the MIL fabric tensor, which describes the orientation of interfaces between bone and marrow.

In the original description of the VO method,[134] a technique for inferring a concentration parameter, κ from individual 2-D sections was developed. There are three prerequisites for using this technique. First, a parametric distribution function must be assumed; second, a geometric model for the trabecular architecture must be assumed; and third, the main orientation of the volume orientation distribution must be known. All of these assumptions are critical and require substantial documentation, and true 3-D measurements evidently are to be preferred.[134]

Some properties of the VO method have been examined. For rectangular particles, the VO method results in blurring around the diagonal orientations,[126,165] but it should be kept in mind that the VO method was not developed for particle systems.

Star volume and star length distribution. The *star volume distribution* (SVD)[27,89,140]—like the volume orientation method—describes the typical distribution of trabecular bone around a typical point in a trabecula (see Figure 2C), and the method was suggested when the VO method was first described.[134] The star volume distribution is closely related to the *star volume* measure.[55,56,176] The *star length distribution* (SLD)[53,140] provides a minor modification of the SVD measure.

The result of SVD (or SLD) measurements may also be expressed by a normalized fabric tensor, the *SVD (SLD) fabric tensor*.[140]

Other Anisotropy Measurements

The anisotropy of curves in 2- and 3-D space may be quantified by counting intersections between the curves and a line or plane, respectively. This has resulted in a method that is closely related to the MIL method,[180] but no evident application exists for cancellous bone. If cancellous bone is skeletonized (**Figure 3**), the method may be used, but the anisotropy of the skeleton will partly be determined by the skeletonization algorithm.[70,104] For granular materials different measures of anisotropy have been suggested[129,130,165] and anisotropy measures do also exist for gray-level images,[167] but no obvious application of these methods exists for cancellous bone.

Mechanical Anisotropy

The quantification of mechanical anisotropy has long been restricted to measuring mechanical properties in an orientation, which has been claimed to be the main orientation, and in one or two orientations perpendicular to this. The validity of the obtained values are affected by minor deviations from the main orientation, which may result in relatively large errors in measured properties.[170] Additional sources of error are the factors mentioned in the *Introduction*.

For sufficiently large homogeneous cancellous bone regions, a method exists which allows full characterization of mechanical anisotropy.[83] The method is, however, not applicable to individual cancellous bone samples.

Snyder et al.[161,162] used a sequential cutting/testing technique to obtain the full compliance matrix. First, cubic specimens with side lengths of 7.8 mm were tested mechanically in compression, and later smaller cubes with side lengths 4.5 mm were cut from the primary cubes. Whereas this approach may be used for homogenous materials,[162] for cancellous bone it is hampered by the effects of structural end phenomena.[90,111,113,132,136] The significance of this problem is indicated by the large fraction of specimens, which had to be excluded from analysis because of inconsistencies.[162]

The availability of 3-D reconstructions made the development of large-scale finite element analysis (LS-FEA) of large cancellous bone regions possible.[75,150,152] The principle of this method is to let the eight corners of a voxel in a 3-D reconstruction form an element in a finite element analysis. The number of elements in an FEA of cancellous bone may, however, reach several hundreds of thousands, which by an ordinary FEA code is inaccessible. By assuming that each individual element has the same mechanical properties, and by using special numerical techniques, the problem may, however, be solved.

The output of an LS-FEA analysis is a full characterization of elastic properties at the cancellous bone continuum level; that is, a full compliance matrix. From this information the mechanical symmetry coordinate system and the mechanical properties in the coordinate system may be determined.[153] One may also calculate local stresses and strains at the trabecular level. This is a significant improvement over previous techniques.

One of the serious problems related to traditional mechanical testing of cancellous bone is the structural end phenomenon, which is caused by impaired mechanical properties of the part of the specimen close to the cut surface.[93,132,136,188] This problem has not been completely eliminated with the LS-FEA technique, but it has been reduced, because the direct mechanics approach used by van Rietbergen et al.[153] averages out the changed mechanical properties close to the specimen surface.

LS-FEA is conceptually closely related to the porous block model of McElhaney.[120,121] In his study from 1970 he stated that "the structural details of porous bone are so complex and variable that an analysis of only a single specimen requires an inordinate effort."[121] The availability of 3-D reconstructions, hardware developments, and new computing techniques have fundamentally changed this situation.

Fabric and Mechanical Anisotropy

To describe differences between the various fabric measurements, Odgaard et al.[140] and van Rietbergen et al.[152] performed studies using cetaceous vertebral cancellous bone. The reason for

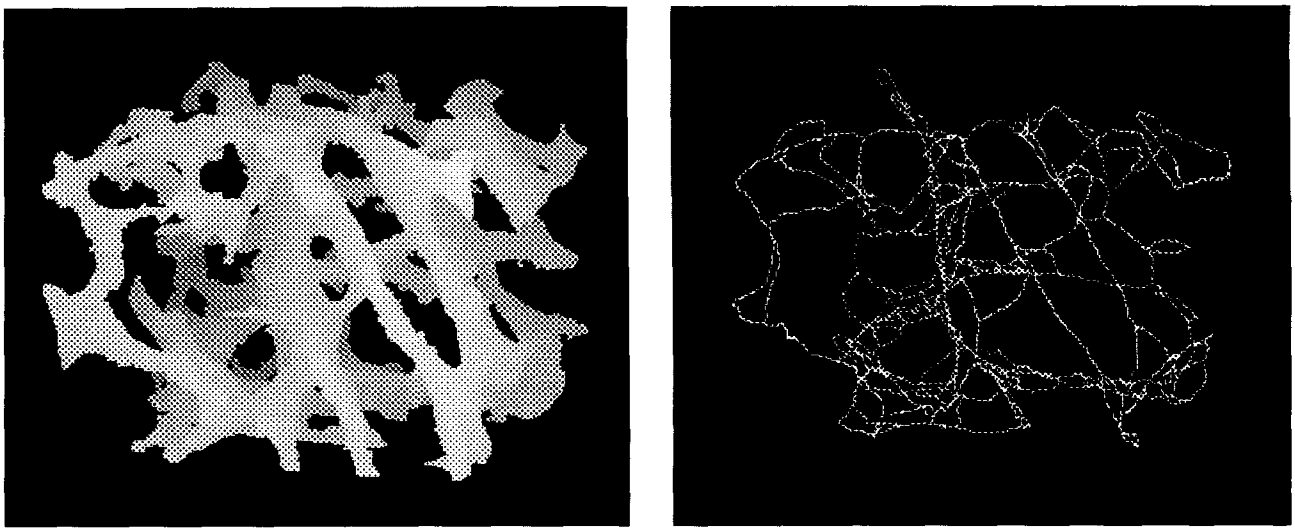

Figure 3. Example of a skeletonization of a cancellous bone cube. The skeletonization has been performed in 3-D space. On the left the original 3-D reconstruction is shown, and on the right the skeletonized set. Removal of any voxel from the skeletonized set will change the topological properties of the set. Connections extending beyond the cube have been eroded.

choosing whale cancellous bone was due to an effort to study a trabecular architecture, which was as homogeneous as possible. In human cancellous bone, there has been shown to be large gradients in architectural properties.[51,52,189] For each of 29 cubic cancellous bone specimens, the fabric tensors (MIL, VO, SVD, SLD) and the LS-FEA calculated compliance matrix were determined. A comparison of anisotropy orientations showed that all architectural measures were highly correlated with FEA-calculated mechanical anisotropy.[140] The primary orientation determined by the architectural measures was within a few degrees of the mechanical primary orientation, but the secondary and tertiary orientations were determined with more uncertainty. This problem was probably caused by relatively transverse isotropy of the specimens studied. Statistical analyses led to the suggestion that SVD is the best predictor of mechanical main orientations.[140]

Only few studies exist that relate mechanical anisotropy to fabric.[51,53,54,63,73,74,152,161,162,172] With two exceptions[53,152] the MIL method has been the only method used to quantify fabric. In some of the studies, several architectural parameters have been included in a multiple regression analysis,[51,73,74] including plate model parameters. In the multiple regression studies, coefficients of determination (r^2) of 0.68–0.90[54] and 0.85–0.87[73,74] for Young's moduli vs. combined measures of density and fabric have been found.

Snyder et al. fitted data to the relations derived by Cowin[21] and found the Young's moduli to be predicted with an r^2 of 0.77, shear moduli with r^2 of 0.98, and Poisson's ratios with r^2 of 0.83.[161] In a later publication, Snyder et al.[162] found only marginal improvements of density–property relations by including MIL anisotropy. The components of the compliance matrix were predicted with r^2 from 0.60 to 0.69, whereas functions of density alone predicted components with r^2 from 0.31 to 0.62. Young's and shear moduli were predicted with r^2 of 0.62–0.75. A similar approach was taken by Turner et al., who found the Young's moduli to be predicted with r^2 of 0.66–0.72 and shear moduli with r^2 of 0.77–0.94.[172]

Residual variation has been explained by architectural inhomogeneity,[51] other important architectural and material properties,[73,172] and methodological errors in determining mechanical properties.[162] The residual variation was partly the reason for introducing the volume-based methods.

van Rietbergen et al.[152] determined the coefficients in Cowin's structure–property relations using both MIL, VO, SVD, and SLD methods, and the coefficients of determination in **Table 1** were found. For all components of the compliance matrix, the best predictability was found using one of the volume-based measures. All r^2 values were very high, and this may be explained by the very homogeneous structure of the cetaceous specimens used. Evidently, these findings will have to be tested in a study using human cancellous bone with larger variability of architectural measures. Such studies are currently being performed.[84]

Table 1. Determination coefficients (r^2) for relations between calculated mechanical properties and fabric

	MIL	VO	SVD	SLD
$\frac{1}{E_i}$	0.968	0.973	0.974	0.980
$-\frac{\nu}{E_i}$	0.924	0.959	0.958	0.957
$\frac{1}{G_i}$	0.967	0.981	0.982	0.982

The terms in the left column refer to entries in the compliance matrix, relating stress to strain in an elastic material; E denotes Young's modulus, ν denotes Poisson's ratio, and G denotes shear modulus. MIL shows consistently lower values than the volume-based measures. From van Rietbergen et al.[152]

Goulet et al.[53] have also presented a comparison of MIL and a volume-based anisotropy measure. They reached the conclusion that both methods predicted mechanical properties equally well, but objections exist to this study. First, a statistical method was not used directly to test differences between predicted anisotropy orientations, and second, not all elements in the compliance matrix were determined.

Specify Fabric Type!

The existence of different fabric measures may seem confusing for those who have thought of architectural anisotropy (fabric) as being a single, well-defined property, but it should be emphasized that a structure may be isotropic with respect to one architectural property and anisotropic with respect to another architectural property. **Figure 4** is a 2-D illustration of this. No method has yet been presented that gives a complete description of the anisotropy of a structure, although combined measures have been described.[165]

The MIL method has, until recently, been the only method available for quantification of fabric, and numerous studies using the method have been published.[24,28,37,40,49,51,52,54,63,68,69,73,74,81,94,105,158,160–162,164,169,171,172,189] The few studies that have provided a thorough comparison of different methods for quantifying fabric indicate, however, that volume-based methods are superior to or at least equally as good as the MIL method.[140,152]

Regardless of which method will finally be accepted as the one to use, one conclusion should remain: There is no single measure that describes every aspect of anisotropy,[72] so any examination of architectural anisotropy should not simply refer to architectural anisotropy, but should explicitly state which property is being examined.

Other 3-D Measurements

Once the 3-D structure of cancellous bone has been represented by a 3-D reconstruction, any parameter pertaining to the architecture can be quantified. The only problem is defining what should be measured. Parameters like trabecular thickness, length, diameter, etc. may intuitively make sense, but a rigorous definition is nontrivial. Only if strict definitions can be given, may these be implemented in algorithms. In this respect 3-D reconstructions challenge our use of architectural terms.

Permeability and conductivity properties of two-phase structures are of much interest in certain fields of materials science.[107] Different methods have been suggested for quantifying these properties. Some methods are pure 2-D methods,[33] which impose some of the restrictions just discussed, other methods are purely qualitative,[144] and yet other methods are 3-D methods.[100,107,166] Some of these techniques may be of relevance in characterizing the architecture of cancellous bone. The *critical sphere*[166] of bone and marrow may, for instance, give an estimate of "thickness" or minimal cross-section of trabeculae and marrow channels.

The dominating assumption about the architectural occurrences during early phases of osteoporosis considers a preferential loss of horizontal trabeculae and a compensatory thickening of vertical trabeculae. Numerous studies have attacked this problem using different 2-D methods.[31,145,159–162] Three-dimensional analysis provides means for conclusions in this mat-

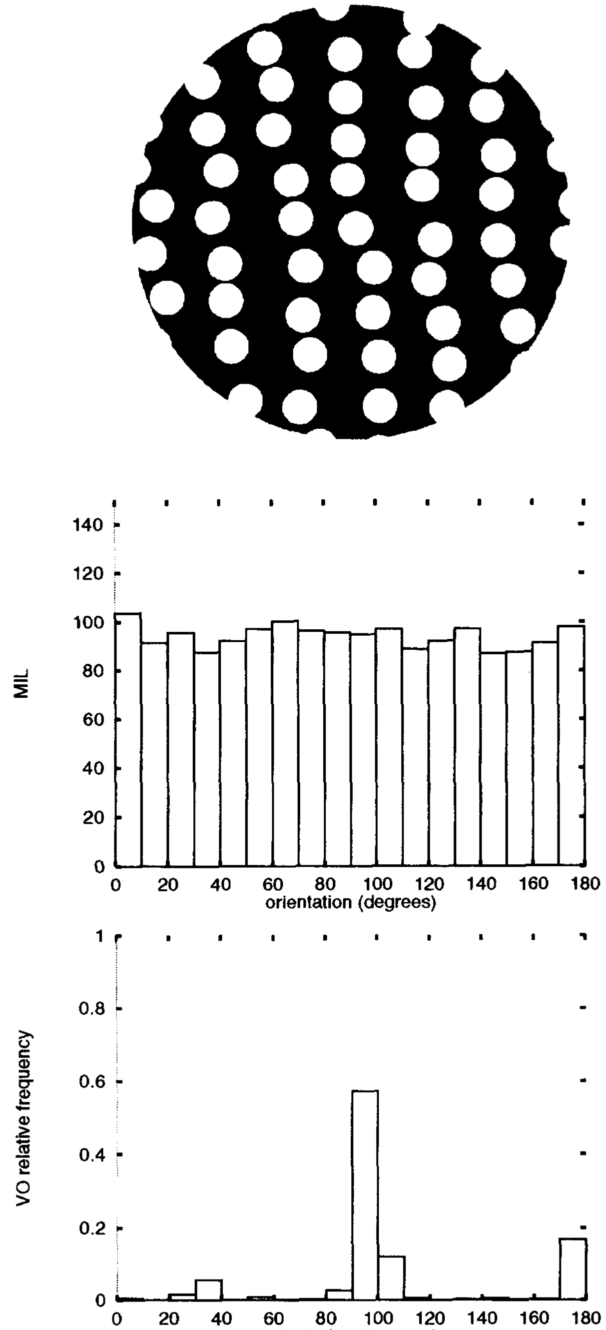

Figure 4. Two-dimensional "Swiss cheese" structure and results of performing mean intercept length (MIL) and volume orientation (VO) measurements. VO measures have been done in the black phase. The MIL ordinate is an arbitrary length measurement, and the VO ordinate is the frequency. Two important effects should be noted. First, the MIL and VO results show dramatic differences, and second, the MIL result is isotropic for the obviously anisotropic structure. The latter effect is caused by the isotropic interface (circles) between phases.

ter. **Figure 5** gives an example of how the volume orientation principle may be used to define the trabecular volume pointing in a specified orientation.

A few investigators have presented results of determining the fractal dimension of cancellous bone.[16,26,181] The fractal dimension value is as yet undetermined. Generally, some publications have no doubt followed from the fractal popularity wave.

Perspectives

The information obtained in 3-D reconstructions may be used to calculate standard histomorphometric quantities. The bone volume fraction, V_V, is easily calculated by dividing the number of bone voxels by the total number of voxels, and the trabecular surface density, S_V, may be calculated using standard stereological or polygon-fitting techniques. Both quantities may, however, easily be obtained using standard stereological techniques on ordinary histological sections,[173,174,180] and one should certainly never start the endeavor of developing a system for 3-D reconstruction just to obtain these quantities.[30] If, on the other hand, a 3-D reconstruction technique is available, then the option of determining standard stereological parameters should be employed.

Various methods known from bone histomorphometry aim at determining connectivity, but none of the conventional methods gives any information about 3-D connectivity. The 3-D reconstruction makes quantification of 3-D connectivity possible, and it does not make much sense to use the 3-D reconstruction for calculations of the conventional histomorphometric variables discussed.

There are physical and chemical limits for the architectural variation in cancellous bone. The consequence of this restriction of architectural variability is that there is probably some relationship between architectural measures, but if the limits change because of some pathological condition, so will the relationship in an unknown manner.

Three-dimensional reconstruction and architectural 3-D analyses provide new means of studying factors, which determine the mechanical properties of cancellous bone. The combination of large-scale finite element modeling and in vivo 3-D studies may disclose how the properties of cancellous bone are maintained and how cancellous bone reacts to changes in mechanical environment. These possibilities are, however, just opening up, and except for density there is not even consensus on which architectural parameters are important to cancellous bone.

Bone Quality

It is generally believed that the cortical shells of most cancellous bone regions contribute only little to the stiffness and strength of the structure, so the interest in the mechanical properties of epi- and metaphyseal regions has concentrated on cancellous bone. There are, however, indications that the significance of the cortical shell may be underestimated.[155,175] Further studies on this subject seem indicated.

A full description of elastic properties is complicated, cancellous bone failure mechanisms are not well understood, and numerous descriptors of cancellous bone architecture have been introduced during the years. These circumstances probably explain why still more bone scientists and clinicians choose to use the term "bone quality" when speaking about bone architecture and mechanics. One should, however, be cautious in using this term uncritically, because an exact terminology does exist. With the new 3-D techniques it is possible to calculate all elastic properties of cancellous bone, and a few 3-D architectural variables seem to provide a replacement for a wealth of older architectural 2-D measures.

In endocrinology and osteoporosis circles there is a long tradition for interest in trabecular connectivity, which is reflected in the large number of surrogate and model-based measures men-

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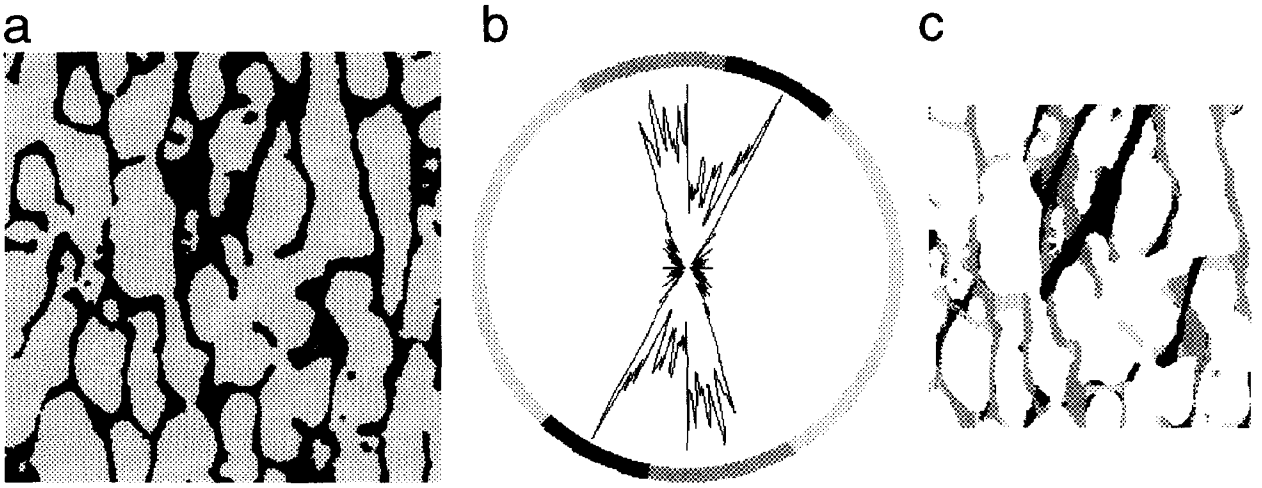

Figure 5. An example of how a directed volume of trabeculae may be determined. (a) The original 2-D section through a cancellous bone cube. (b) Polar plot of local volume orientation frequencies for the 2-D structure. The circle around the polar plot identifies populations of local volume orientations and marks these by different shades of gray. (c) Each pixel in the original 2-D section is given a gray-level value corresponding to the local volume orientation given the classification in (b). Note that a guard area excludes determinations of local volume orientation close to the artificial borders of the original 2-D section.

tioned previously, but this is in sharp contrast to the situation in biomechanics circles. Biomechanicians have not seriously considered connectivity, but have taken much interest in density and anisotropy, and in Cowin's constitutive fabric–mechanics relations these two measures are the only architectural variables.

The latter view finds support from both theoretical and empirical studies. First, theoretical studies of open-celled structures indicate that the number of trabeculae is of no importance for the mechanical properties.[3,45,46] Second, new studies indicate that more than 90% of the variance in mechanical properties—and these are extensive descriptions of mechanical properties—may be explained by density and anisotropy alone,[140,152] in which case there is no need for further architectural variables. However, at least one study based on model simulations indicates that connectivity may indeed have important effects on mechanical properties.[36] Only few experimental studies have addressed the effect of connectivity on mechanical properties,[49,50,73,74] but for reasons already discussed, there certainly is a need for further studies in this area.

At the continuum level of cancellous bone, many remodeling studies of density (inner geometry) and form (outer geometry) have been performed using finite element methods. Similar techniques may be imagined at a microarchitectural level,[94,128] but in this case the verification of methods seems even more difficult than for the continuum case.

Back to 2-D?

With the advent of 3-D reconstructions, it has become possible to study architectural properties, which have hitherto been inaccessible. Loosely used terms like "trabecular number" and "trabecular orientation" have been given precise definitions thanks to three-dimensional thinking. Other terms from traditional bone histomorphometry should probably also be revised, and one should try to give precise and model-free 3-D definitions of these parameters.

Studies currently performed by different centers may result in the consensus that cancellous bone mechanical properties are adequately described by density and fabric, and, should this be the case, SVD seems to be the most likely fabric measure. Connectivity may, however, also be found to be an important predictor of mechanical properties. Two-dimensional determination of SVD is possible with the use of vertical sections,[89] and the ConnEulor[57] provides an unbiased connectivity measure using only pairs of 2-D sections. Consequently, the architectural features of importance for mechanical properties may be accessible with 2-D sections.

The large-scale finite element analysis, which has advantages over traditional mechanical testing, does, however, require 3-D reconstructions. On a long-term basis, the artifact-prone traditional mechanical testing may for several purposes be replaced by LS-FEA.[151] Future detailed in vivo studies and experimental in vitro studies are also likely to use 3-D imaging, so a wealth of 3-D studies can be expected in the future.

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